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A Multimodal Approach to Global Geolocation: Fusing Vision Backbones with Temporal Metadata

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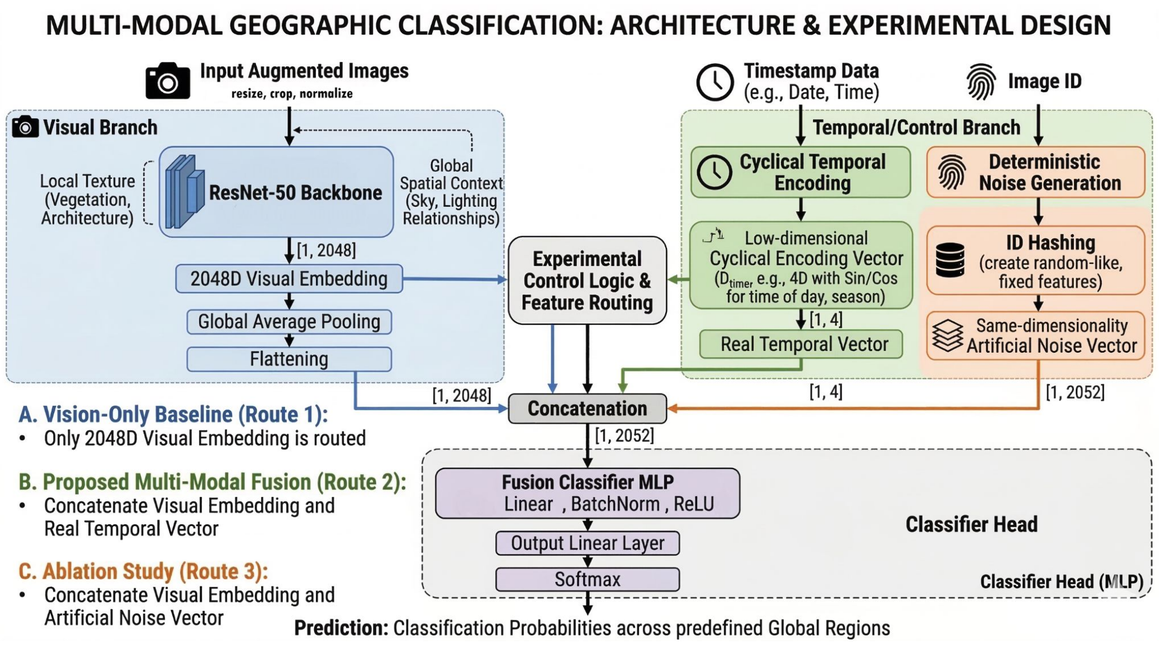
Introduction & Research Question

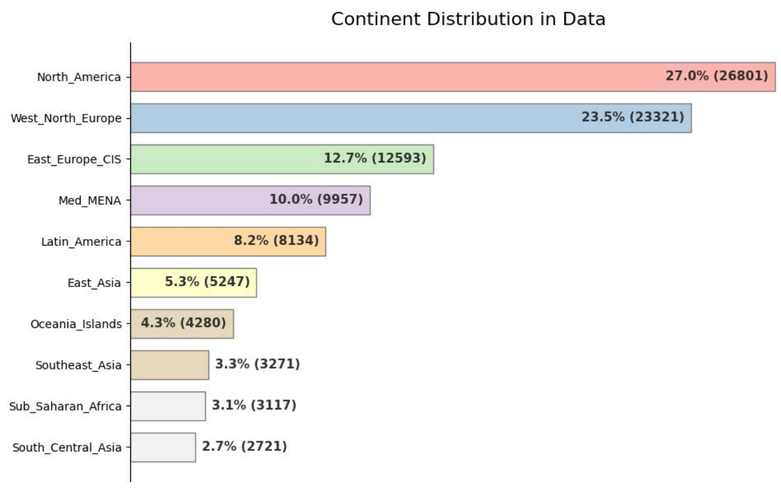
- **The Task:** Predicting the broad geographic location (framing geolocation as a robust classification task across 10 major global regions) of an image based on its visual contents and metadata.
- **The Research Question:** To what extent does explicitly integrating temporal metadata (season, time of day) with visual features improve the regional classification accuracy compared to a vision-only baseline?
- **Hypothesis:** Knowing *when* an image was taken acts as a vital constraint that helps the model better predict *where* it was taken by resolving hemispherically ambiguous visual features.



Dataset& Preprocessing

We utilize the **OpenStreetView-5M (OSV-5M)** dataset, implementing specific strategies to ensure robust evaluation:

- **Geographic Aggregation:** To focus on macro-climate differences, we map over 225 individual country codes into a dozen of **distinct global macro-regions**. This reframes the task into a balanced, manageable classification problem.
- **Visual Preprocessing:** Images are dynamically resized to 224×224 using random resized cropping for data augmentation.
- **Temporal Preprocessing:** Timestamps are transformed via cyclical encoding (sine/cosine mapping) to preserve periodic temporal relationships(e.g., ensuring 11:59 PM and 0:01 AM are mathematically proximate).



References

[1] Astruc, G., et al. "OpenStreetView-5M: The Many Roads to Global Visual Geolocation." CVPR (2024)
[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun; CVPR, 2016, pp. 770-778

Motivation

Relying solely on pixels causes severe geographical ambiguity:

- **Seasonal Inversion:** Snow implies North in Jan, but South in Jul, which causes ambiguity.
- **Lighting Ambiguity:** Shadow directions are unresolvable without time-of-day.
- **Solution:** Explicit time data decouples temporal variations from genuine geographic features.

Methodology

A. Vision-Only Baseline:

- Augmented images are used to train a standard randomly initialized ResNet-50 backbone. The network relies solely on extracted visual features to extract local textures (e.g., vegetation, architecture) and captures global spatial context (e.g., sky and lighting relationships).

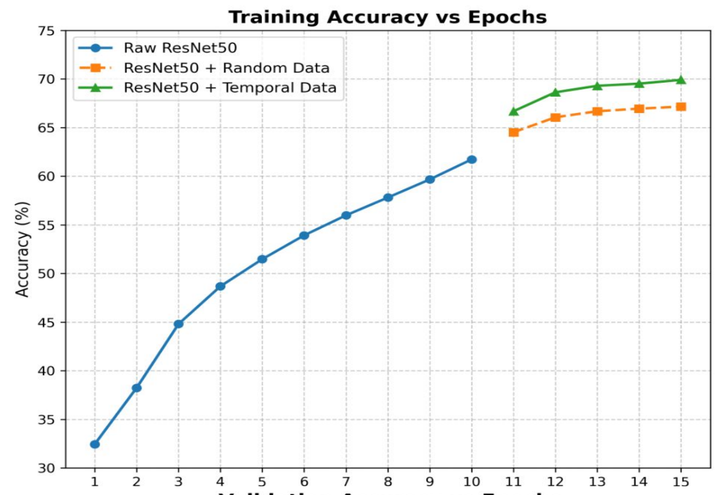
B. Proposed Multi-Modal Fusion:

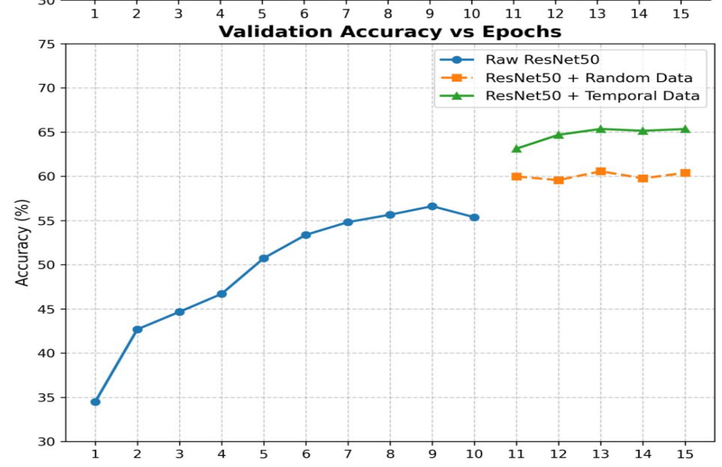
- **Visual Branch:** Baseline ResNet-50 above.
- **Temporal Branch:** Timestamps are converted into a low-dimensional cyclical encoding vector.
- **Integration:** The visual embedding and the cyclical temporal vector are directly concatenated. This joint representation is then processed by an MLP classification head.

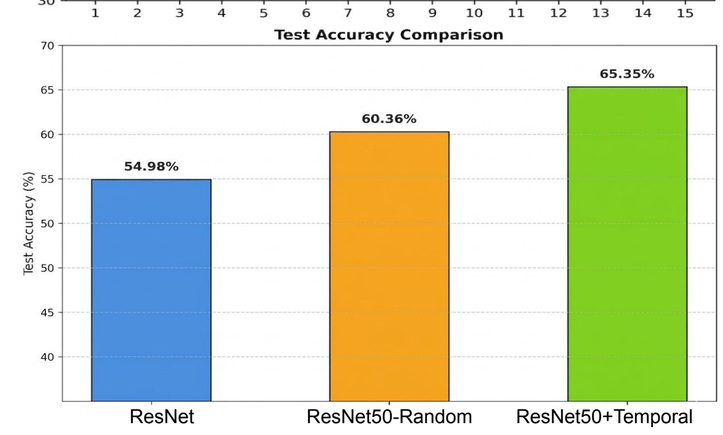
C. Ablation Study:

- Actual timestamps are intentionally replaced with random noise vectors of identical dimensionality. These features are deterministically generated by hashing the unique image IDs, which is irrelevant in this context.

Results







Fusion with temporal data consistently yields **higher accuracy** during both training and validation, demonstrating that the enhanced performance originates from the **temporal semantics** rather than mere architectural modifications. This consistent superiority is robustly reflected in the final testing accuracy as well.

Future Work

- Data Augmentation
- Hemispheric Balancing
- Compare current Early Fusion vs. Late Fusion
- Interpretability: Visualize how temporal information enhances model performance